Phase 0 Decision Record

This record is intentionally incomplete. It lists the decisions that must be verified before monitoring or cloud deployment work begins.

Confirmed scope

- MCPulse monitors remotely reachable MCP servers.
- Local stdio-only monitoring and AI-client tool-call interception are out of scope.
- AWS provisioning requires explicit approval after Learner Lab access is verified.

MCP test server decision

Selected target: DeepWiki public MCP server

Field	Confirmed value
Endpoint	`https://mcp.deepwiki.com/mcp`
Provider documentation	`https://mcp.deepwiki.com/`
Transport	Streamable HTTP
Authentication	None documented for public repositories; private-repository access uses a separate endpoint and bearer token
Verified on	2026-09-27
Initialization result	HTTP 200; `text/event-stream`; MCP protocol version `2025-03-26`; server name `DeepWiki`, version `2.14.3`
Tool discovery result	HTTP 200; three public tools returned: `ask_wiki_question`, `read_wiki_contents`, and `read_wiki_structure`

The verification used bounded `initialize` and `tools/list` JSON-RPC requests only. It did not call a tool, send credentials, or persist a session. The monitoring engine should first run `initialize` as the protocol health check, measure its full request duration, and optionally run `tools/list` as metadata discovery. It must accept both JSON and request-scoped SSE responses.

Rejected target: OpenMCP Korea Weather

The documented endpoint `https://mcp.open-mcp.org/api/server/korea-weather@latest/mcp` was reachable on 2026-09-27 but returned HTTP 500 with the JSON-RPC error `Failed to initialize McpServer instance.` It is not suitable as the initial healthy demo target. It can inform an outage/failure test only if its behavior is rechecked at that time.

Pending decisions

Decision	Status	Required before
Remote MCP endpoint and transport	Confirmed: DeepWiki public Streamable HTTP	MCP checker
Exact timeout, retry and accepted protocol-version policy	Unconfirmed	MCP checker
Endpoint authentication and secret handling	Unconfirmed	MCP checker
AWS Learner Lab services, regions, budget and session limits	Unconfirmed	Deployment work
Deployment topology	Unconfirmed	Deployment work
Management-route authentication for the demo	Unconfirmed	Server registry API
Default interval, timeout, degraded threshold, failure threshold and recovery rule	Unconfirmed	Monitoring engine

Monitoring rules selected for the local prototype

Setting	Value
Check protocol	Streamable HTTP `initialize`, using protocol version `2025-03-26`
Check timeout	10 seconds (`CHECK_TIMEOUT_MS`)
Default interval	60 seconds; allowed range is 30-3600 seconds
Scheduler evaluation	Every 10 seconds
Incident threshold	3 consecutive failures (`INCIDENT_FAILURE_THRESHOLD`)
Recovery rule	First successful subsequent check resolves any open incident
Slow-response status	Not implemented yet; response duration is recorded but no separate threshold is configured

The manual-check management route is local development only until the authentication approach is selected. It must not be publicly exposed in a deployment.